VAMSIKRISHNA CHINNAM

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EDUCATION

University of Southern California **Los Angeles, CA**
Master of Science in Computer Science (GPA: 3.9/4.0) *Graduation Date: May 2026*
Coursework: Data Structures & Algorithms, Machine Learning, NLP, Information Retrieval, Database Systems, Security Systems

WORK EXPERIENCE

BreatheIt **Los Angeles, CA**
Software Engineer Intern *June 2025 – September 2025*

- Owned end-to-end development of a cloud-backed scheduling platform (React, TypeScript, React Native, Node.js); designed APIs and data models, integrated Stripe and Google Calendar, and supported production releases via CI/CD.
- Designed and operated scalable backend services on GCP for LLM-powered decision workflows, focusing on latency, fault tolerance, monitoring, and safe production rollout, with real-time inference and guardrails for reliability.

Indian Space Research Organization (ISRO) **Andhra Pradesh, IN**
Software Engineer Intern *March 2024 – June 2024*

- Developed and deployed an internal React.js web platform within ISRO's secured environment, leveraging Chakra UI and Tailwind CSS to deliver intuitive interfaces for ticketing, communication, and role-based workflow orchestration.
- Built large-scale ML pipelines using Vision Transformers on satellite imagery (Sentinel-1 SAR), handling noisy real-world data, evaluation, and deployment constraints; improved model accuracy by 15% through architecture and training optimizations.

Suvidha Foundation **Chennai, IN**
Python Developer Intern *December 2021 – February 2022*

- Constructed scalable Python data pipelines (ETL Workflows) for ingestion, validation, and transformation using NumPy, Pandas, and PyTorch, designed to handle failures and inconsistent inputs.
- Improved system reliability and throughput by optimizing pipeline execution paths, batching, and error-handling logic, increasing throughput by 25% and reducing pipeline failures by 15% in production workloads.

The Sparks Foundation **Chennai, IN**
Web Developer Intern *July 2021 – August 2021*

- Implemented a full-stack banking web application using React.js, JavaScript, PHP, and MySQL, supporting real-time balance updates, transaction history, and secure fund transfers via RESTful APIs.
- Built and optimized backend services for account management, authentication, and transaction processing, ensuring data integrity, reliable API communication, and a responsive user experience.

PROJECTS & PUBLICATIONS

AI Interview Coach (Python, LLaMA, Gemini API, React, FastAPI) ***GitHub***

- Built an AI-driven interview preparation platform, designing and deploying LLM-powered backend services (T5 fine-tuning, LLaMA, Gemini API) for question generation and evaluation, with a React + FastAPI interface to deliver real-time, user-facing workflows.

Scalable File Storage and Sharing Backend (MongoDB, Node.js, Express, AWS S3) ***GitHub***

- Architected a distributed backend service separating metadata in MongoDB from objects on AWS S3; designed scalable APIs, asynchronous processing workflows, and data consistency guarantees for durable uploads, downloads, and sharing.

Full-Stack Booking & User Management System (HTML, CSS, Node.js/Express, SQL, PHP/DBMS) ***GitHub***

- Developed a secure hotel booking system with an adaptive frontend and session hardening (HttpOnly, Secure, SameSite); built Node.js backend with transaction-backed MySQL and optimized queries for reliability.

Vehicle Trajectory Mining and Traffic Prediction using Object Detection (Python, TensorFlow, OpenCV) ***Publication Link***

- Built a real-time multi-camera traffic system using YOLO and DeepSORT on OpenCV to track across geolocated CCTV feeds, with LSTM-based congestion forecasting; shipped Streamlit/Plotly Dash dashboards (matplotlib, seaborn) over 10,000 frames, achieved 10% higher accuracy than PrefixSpan, and was validated by Chennai Police (65% effective).

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, Scala | **Backend & Web**: Node.js, React, React Native, Angular, FastAPI, REST APIs, HTML, CSS | **Cloud & Infrastructure**: AWS EC2, AWS S3, GCP, Docker, Kubernetes, CI/CD | **Data**: MySQL, MongoDB, Hadoop, Firebase | **Testing**: Unit Testing, Integration Testing, Selenium | **ML Libraries:** PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Plotly, spaCy, OpenCV, CuDF, Streamlit